

Objective of the Task

This week's task is divided into two parts. In the first part, you will investigate the steel plant energy dataset by performing a deep exploratory data analysis, engineering new features from existing columns, and identifying patterns and data quality issues that could affect model training. In the second part, you will use the engineered dataset to train multiple regression models, evaluate them properly, and establish a solid baseline by documenting which model performs best and why. Together, these two parts represent the core workflow of a real machine learning project.

Dataset

- **Name:** Steel Industry Energy Consumption Dataset
- **Link:** <https://archive.ics.uci.edu/static/public/851/steel+industry+energy+consumption.zip>
- **Description:** Real energy consumption data from a steel manufacturing plant, containing usage readings, load types, power factors, and timestamps.

Part 1 — Deep Exploratory Analysis & Feature Engineering

Requirements

1. Load the dataset and re-examine its structure.
2. Convert the date column to datetime and extract: hour of day, day of week, month, and whether the day is a weekday or weekend — add these as new columns.
3. Create a new feature called `Power_Factor_Ratio` by dividing leading power factor by lagging power factor.
4. Create a new binary feature called `High_Load` by marking rows where `Usage_kWh` is above the 75th percentile as 1 and the rest as 0.
5. Detect outliers in `Usage_kWh` using the IQR method — print how many outliers exist and visualize them with a boxplot.
6. Plot a correlation heatmap of all numerical features and identify the top 3 features most correlated with `Usage_kWh`.

7. Create a grouped bar chart showing average energy consumption by Load Type (Light Load, Medium Load, Maximum Load).
8. Create a line chart showing average energy usage by hour of day.
9. Write a 250-word EDA Summary at the end covering: data quality issues found, top correlated features, the most interesting pattern discovered, and one hypothesis about what drives energy spikes.

Part 2 — Baseline Regression Modeling

Requirements

1. Load the dataset with all engineered features from Part 1.
2. Drop the original date column and any columns that directly leak the target (such as `High_Load`).
3. Encode categorical columns like `Load_Type` and `Day_of_week` using one-hot encoding or label encoding — document which you chose and why.
4. Split the data 80% training and 20% testing using `random_state=42` for reproducibility.
5. Train all 4 models: Linear Regression, Ridge Regression, Decision Tree Regressor, and Random Forest Regressor.
6. For each model, calculate and print MAE, RMSE, and R-squared on the test set.
7. Run 5-fold cross-validation on each model and report the mean RMSE across folds.
8. Create a bar chart comparing the test RMSE of all 4 models side by side with proper labels and title.
9. For your best model, plot a scatter chart of Predicted vs Actual values.
10. Write a Model Selection section in markdown explaining which model performed best, the cross-validation RMSE, any signs of overfitting, and which model you will carry forward.

